

LINEAR ALGEBRA TEST

For each of the following matrices compute &

- the rank
- the number of linearly independent rows
- the number of linearly independent columns

Moreover, tell if the linearly independent rows/columns form a basis of $\mathbb{R}^3$

1. $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

2. $\begin{bmatrix} 2 & 3 & 4 \\ 1 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

3. $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & -1 \end{bmatrix}$

4. $\begin{bmatrix} -1 & 2 & 3 \\ -1 & 2 & 3 \\ -2 & 4 & 6 \end{bmatrix}$

5. $\begin{bmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

1. $\det = 1 \neq 0 \Rightarrow \text{max rank} = \text{rank} = 3$

rows are lin indep

columns are lin indep

$\Rightarrow$ they form a basis for $\mathbb{R}^3$

2. One row it's equal 0!

Cannot have maximum rank $\Rightarrow$ rows and columns

cannot form a basis

There are 2 lin indep rows (the central column guarantees it)

There are 2 lin indep columns (the third one can be eliminated, it's a multiple of the first one)

CHECK WITH GAUSSIAN ELIMINATION!

3. $\det = -6 \Rightarrow$ max rank
 $\Rightarrow$ rows are lin indep
columns are lin indep
 $\Rightarrow$ they form a basis

4. row 2 is equal to row 1
row 3 is a multiple of row 1
 $\Rightarrow$ rank = 1
rows and columns cannot form a basis

5. $\det = 2 \Rightarrow$ max rank
 $\Rightarrow$ rows and columns are lin indep
 $\Rightarrow$ they form a basis

USE GAUSSIAN ELIMINATION TO REDUCE THE FOLLOWING MATRICES
to ECHELON FORM (i.e., the STAIRCASE PATTERN)

$$1. \begin{bmatrix} 1 & 3 & 4 & -1 \\ 2 & -1 & 5 & 1 \\ 3 & 2 & 0 & 4 \end{bmatrix} \quad R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 1 & 3 & 4 & -1 \\ 0 & -7 & -3 & 3 \\ 3 & 2 & 0 & 4 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1 \quad \begin{bmatrix} 1 & 3 & 4 & -1 \\ 0 & -7 & -3 & 3 \\ 0 & -7 & -12 & 7 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \quad \begin{bmatrix} 1 & 3 & 4 & -1 \\ 0 & -7 & -3 & 3 \\ 0 & 0 & -9 & 4 \end{bmatrix}$$

$$R_2 \rightarrow -\frac{1}{7}R_2 \quad \begin{bmatrix} 1 & 3 & 4 & -1 \\ 0 & 1 & 3/7 & -3/7 \\ 0 & 0 & -9 & 4 \end{bmatrix}$$

$$R_3 \rightarrow -\frac{1}{9}R_3 \quad \begin{bmatrix} 1 & 3 & 4 & -1 \\ 0 & 1 & 3/7 & -3/7 \\ 0 & 0 & 1 & -4/9 \end{bmatrix}$$

$$2. \begin{bmatrix} 2 & -1 & 1 \\ 1 & 1 & -1 \\ 3 & 1 & -2 \end{bmatrix} \quad R_2 \rightarrow R_2 - \frac{1}{2}R_1 \quad \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3/2 & -3/2 \\ 3 & 1 & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - \frac{3}{2}R_2 \quad \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3/2 & -3/2 \\ 0 & 5/2 & -7/2 \end{bmatrix}$$

$$R_2 \rightarrow \frac{2}{3}R_2 \quad \begin{bmatrix} 2 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 5/2 & -7/2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - \frac{5}{2}R_2 \quad \begin{bmatrix} 2 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_1 \rightarrow \frac{1}{2}R_1 \quad \begin{bmatrix} 1 & -1/2 & 1/2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$3. \begin{bmatrix} 1 & 0 & 2 \\ 2 & 3 & 4 \\ 0 & 0 & 0 \end{bmatrix} \quad R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \rightarrow \frac{1}{3}R_2 \quad \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$4. \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \quad R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ -2 & 7 & 2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_1 \quad \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 8 & 3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2 \quad \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_1 \rightarrow \frac{1}{2}R_1 \quad \begin{bmatrix} 1 & 1/2 & 1/2 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow -\frac{1}{8}R_2 \quad \begin{bmatrix} 1 & 1/2 & 1/2 \\ 0 & 1 & 1/4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$5. \begin{bmatrix} 1 & 3 & 4 \\ 2 & 5 & 8 \\ 3 & 7 & 12 \end{bmatrix} \quad R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 1 & 3 & 4 \\ 0 & -1 & 0 \\ 3 & 7 & 12 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1 \quad \begin{bmatrix} 1 & 3 & 4 \\ 0 & -1 & 0 \\ 0 & -2 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 2R_2 \quad \begin{bmatrix} 1 & 3 & 4 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \rightarrow -R_2 \quad \begin{bmatrix} 1 & 3 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$